Vision-Language Grounded Semantic Navigation for Autonomous Mobile Robots

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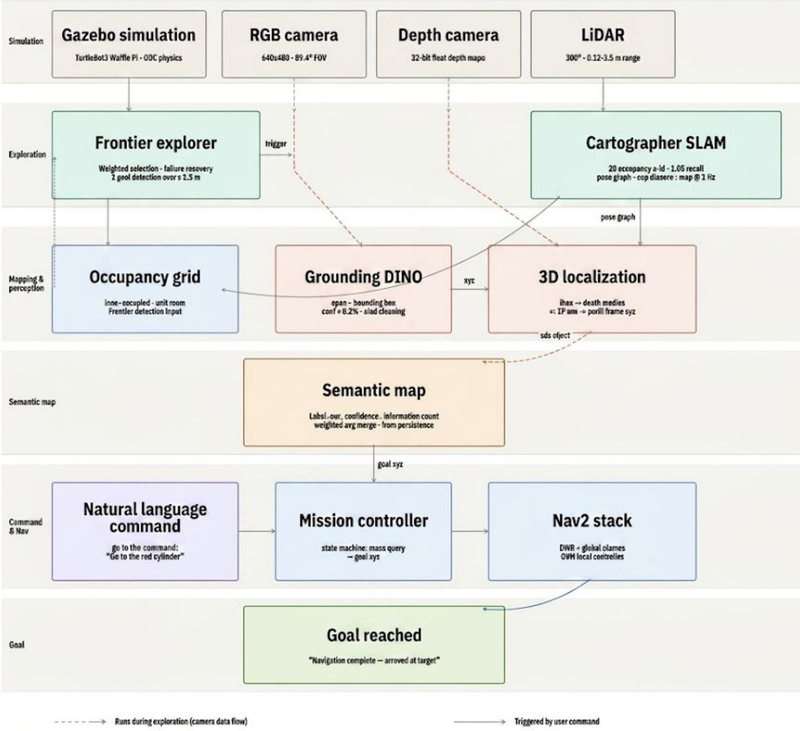
Overview

- **The Gap:** Traditional robotic navigation relies on coordinate-based goals (X, Y), which are unintuitive for users and assume pre-built maps with known object labels.
- **The Goal:** Enable robots to interpret compositional natural language commands (e.g., "go to the red cylinder").
- **Key Challenge:** Bridging the semantic gap, i.e, translating high-level language instructions into low-level metric navigation goals.

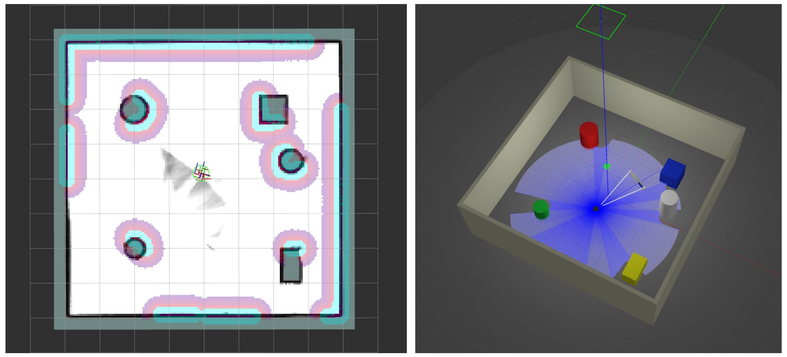
Approach



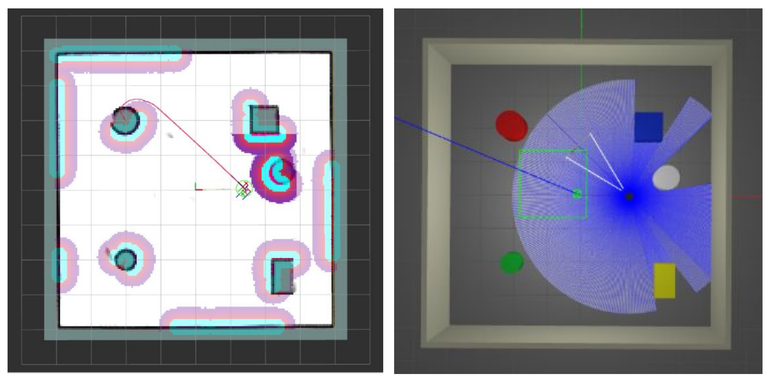
- **Environment:** Indoor Gazebo simulation with a single room containing multiple geometric objects.
- **Hardware Setup:** TurtleBot3 Waffle Pi equipped with a 2D LiDAR and RGB Camera
- **Testing Suite:** Multiple systematically generated language commands (including valid and invalid queries)

Methodology

- **Semantic Mapping:** The robot explores the environment using SLAM to build a geometric map. During exploration, Grounding DINO detects objects from camera images. These detections are projected into 3D using depth and transformed into the global frame. The system stores object labels and positions in a semantic map, enabling navigation based on natural language queries.



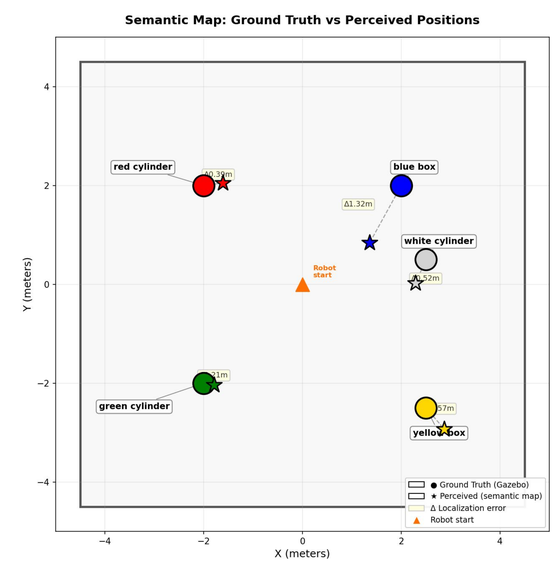
- **Vision-Guided Planning:** User commands are parsed to extract the target object, which is matched against the semantic map to obtain its location. The system then generates a navigation goal and uses Nav2 to plan and execute a path to the target.



```
[mission_controller_node-19] [INFO] [1776449709.667984963] [mission_controller_node]: Received command: 'Go to red cylinder'
[mission_controller_node-19] [INFO] [1776449709.669612662] [mission_controller_node]: Navigating to 'red cylinder' at [-1.61, 2.05]
[bt_navigator-12] [INFO] [1776449709.673659572] [bt_navigator]: Begin navigating from current location (-1.18, 2.20) to (-1.61, 2.05)
[mission_controller_node-19] [INFO] [1776449709.675862161] [mission_controller_node]: Navigation goal accepted
[controller_server-8] [INFO] [1776449709.697030767] [controller_server]: Received a goal, begin computing control effort.
[controller_server-8] [INFO] [1776449709.704170301] [controller_server]: Reached the goal!
[bt_navigator-12] [INFO] [1776449709.735402675] [bt_navigator]: Goal succeeded
```

Simulation Results

- **Overall Success:** On average, the system achieves ~58% across a robustness evaluation including negative and ambiguous queries, and 100% success on standard navigation commands.
- **Localization Accuracy:** Typical error is ~0.2–2.5 m, with occasional larger deviations under perception and mapping failures.



- **Failure Modes:** Errors arise from limited spatial reasoning, vocabulary mismatches, and localization noise. Additional issues include detection noise, map incompleteness, and ambiguity between similar objects. The system assumes a static environment.

Conclusions & Future Scope

- **Takeaways:** Integrating open-vocabulary vision-language models (e.g., Grounding DINO) with ROS 2 enables intuitive human-robot interaction without requiring fixed object categories.
- **Future Work:** Improve spatial reasoning for complex queries and enhance SLAM for more accurate and consistent mapping. Extend the system to dynamic environments and deploy it on physical TurtleBot3 hardware.